FNB APP OF THE YEAR HACKATHON 2026 | ACCESS TO HIGHER EDUCATION | TEAM THUTO

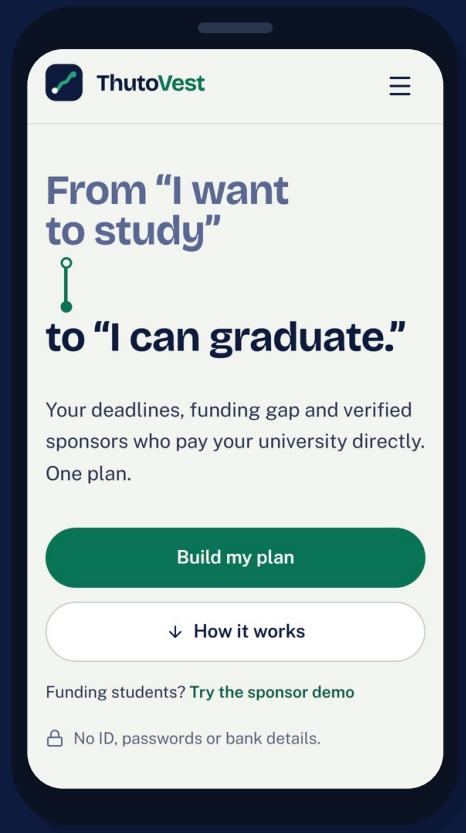
ThutoVest

From “I want to study”

to “I can graduate.”

A personal plan, early warnings and verified sponsors who pay the university directly. Free for students.

Thuto means learning in Setswana and Sesotho. *Vest* means investing in a student.



Scan to try it
thutovest.vercel.app

BUILT BY

Samalani Maluwa
Reshoketswe Makhura
Tebogo Sasa

WHY

Matriculants plan their future alone

The challenge: help young South Africans choose a path, fund it, apply and stay supported until graduation.

893,847
NSFAS funding applications for 2026

218,000
still incomplete because documents were missing

Scattered information
A plan pieced together from NSFAS, university sites and WhatsApp groups.

Missed deadlines
Nothing warns them in time, so places and funding slip away.

The missing middle
Households on R350,000 to R600,000 earn too much for NSFAS, but can't pay fees.

Bursary scams
Fake offers ask for "admin fees", and students can't tell real from fake.

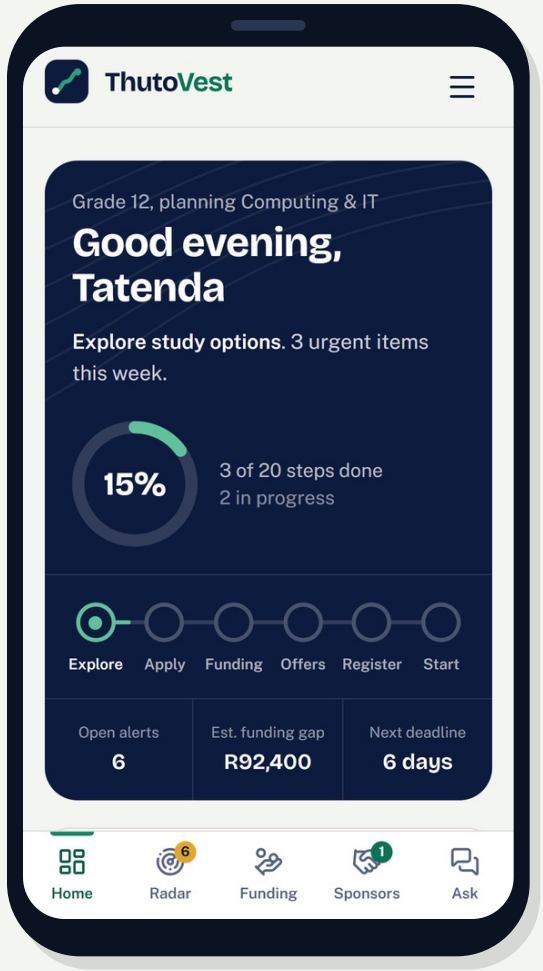
Funded, then dropped
Marks slip quietly, and students hear about it when the funding is gone.

Sources: NSFAS board chair, reported by TimesLive and EWN, 6 January 2026. Missing middle: households earning R350,000 to R600,000 a year. Scams: Daily Maverick, 8 April 2025.

MEET OUR STUDENT

Tatenda, 17, wants to study IT at UJ

Grade 12 in Gauteng. She will be the first in her family at university.



APS 32
Enough for IT at UJ, if her marks hold

R350k to R600k
Her household income: the missing middle

R116,900
One year at UJ, with residence

R24,500
What her family and savings can put in

One phone
Little data, so WhatsApp matters

Under 18
Her mother Nomsa consents and co-signs

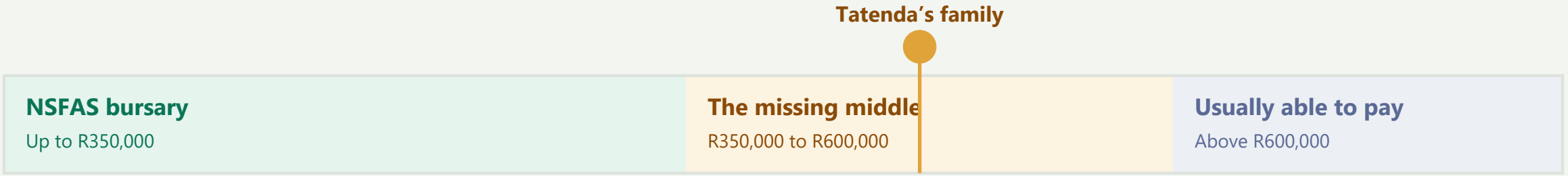
Still to find for her first year: R92,400

Tatenda is a fictional demo student; the figures are from her plan in the app.

WHY TATENDA FALLS THROUGH THE GAP

Too much for NSFAS. Not enough for fees.

Household income a year, and where Tatenda's family sits.



R116,900
One year at UJ: tuition R58,200, residence R42,000, books, transport and other costs

–

R24,500
What her family and holiday savings can put in

=

R92,400
The gap no public bursary covers for her

NSFAS bursary limit: R350,000 combined household income a year (R600,000 for students with a disability). ThutoVest connects students like Tatenda with verified sponsors who fund the gap.

WHAT THUTOVEST IS

One route from matric to graduation

A free app that plans the way, finds the funding, and keeps students funded once they are studying.



Plan

A personal plan with one next best action, deadlines, documents and the funding gap in rands.

Fund

Verified sponsors find the student and pay the institution directly. Never the student, never a fee.

Stay

Marks read after every test, early help, and fair warnings, so funding is not lost out of nowhere.

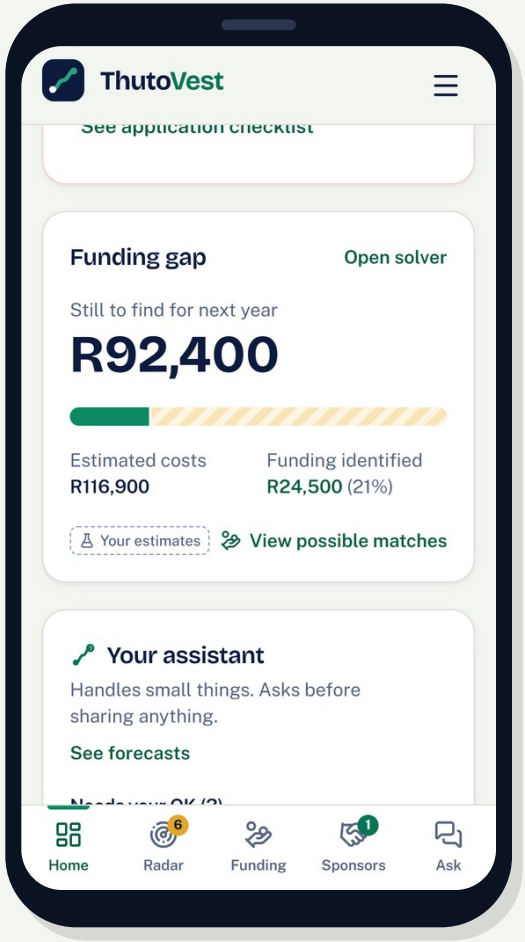
Free for students, always. Sponsors pay a small admin fee on awards paid through the platform.

STOP 1: PLAN

A plan that tells her what to do next

Built from a few questions, and it re-plans itself when her situation changes.

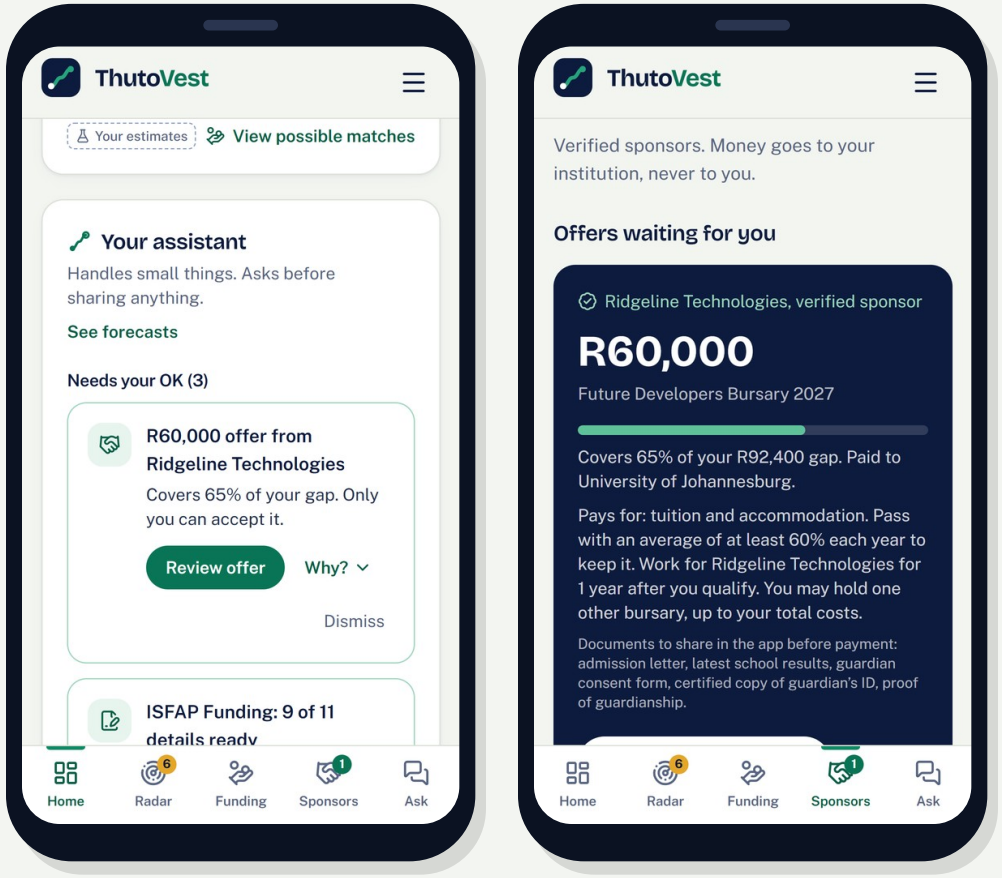
- **One next best action**, with the date and why it matters: “Submit your first-choice application, 6 days left”.
- **Risk Radar** warns before a deadline, a missing document or a money gap becomes a problem.
- **66 qualifications** in 8 fields, re-scored the moment her APS changes, including diplomas and higher certificates.
- **All 26 public universities and 50 TVET colleges**, with official application, funding and residence links.
- **The funding gap in rands**: R92,400, and which aid can close it.
- **WhatsApp**: Talk to ThutoVest when data is low.



STOP 2: FUND

A verified sponsor finds her

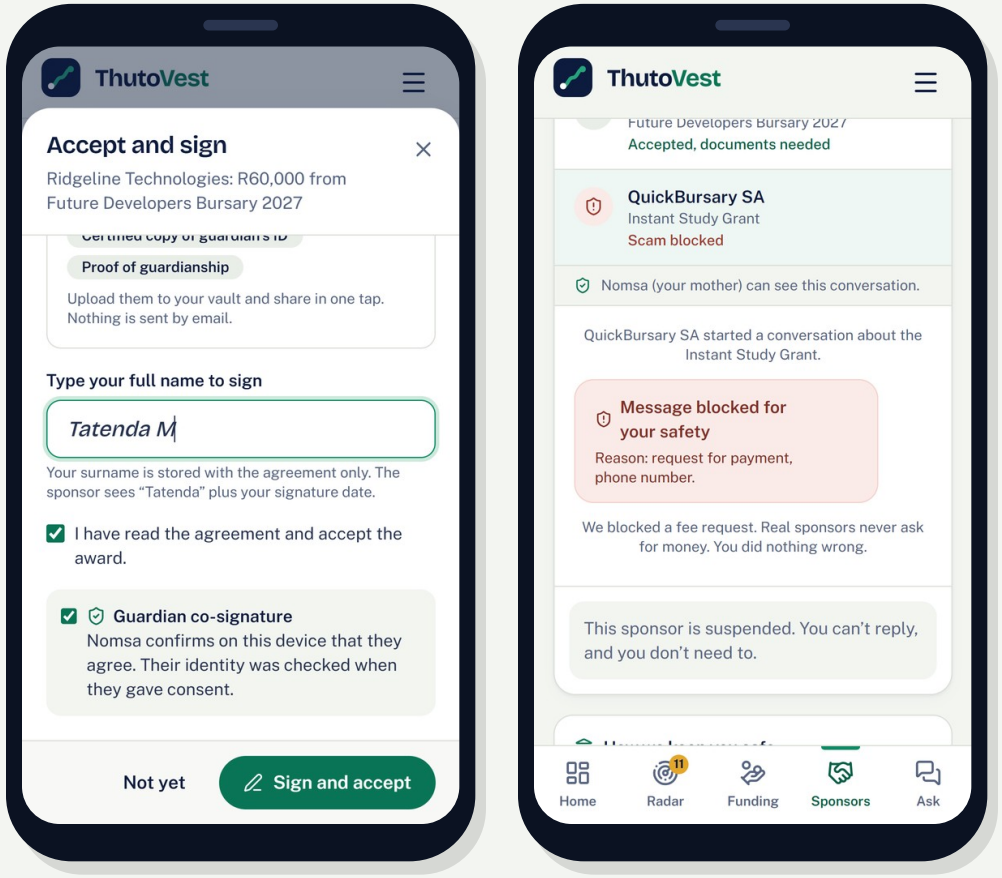
- Sponsors see an **anonymised, verified profile**: first name, field, province, gap and badges. Never her ID or surname.
- Ridgeline offers **R60,000** for tuition and residence. It reaches her phone in seconds.
- No offer can go **above the gap**, and a student holds at most two bursaries.
- The money goes **straight to the university**. R32,400 is left for a second bursary.



STOP 3: PROTECT

Children protected, scams stopped

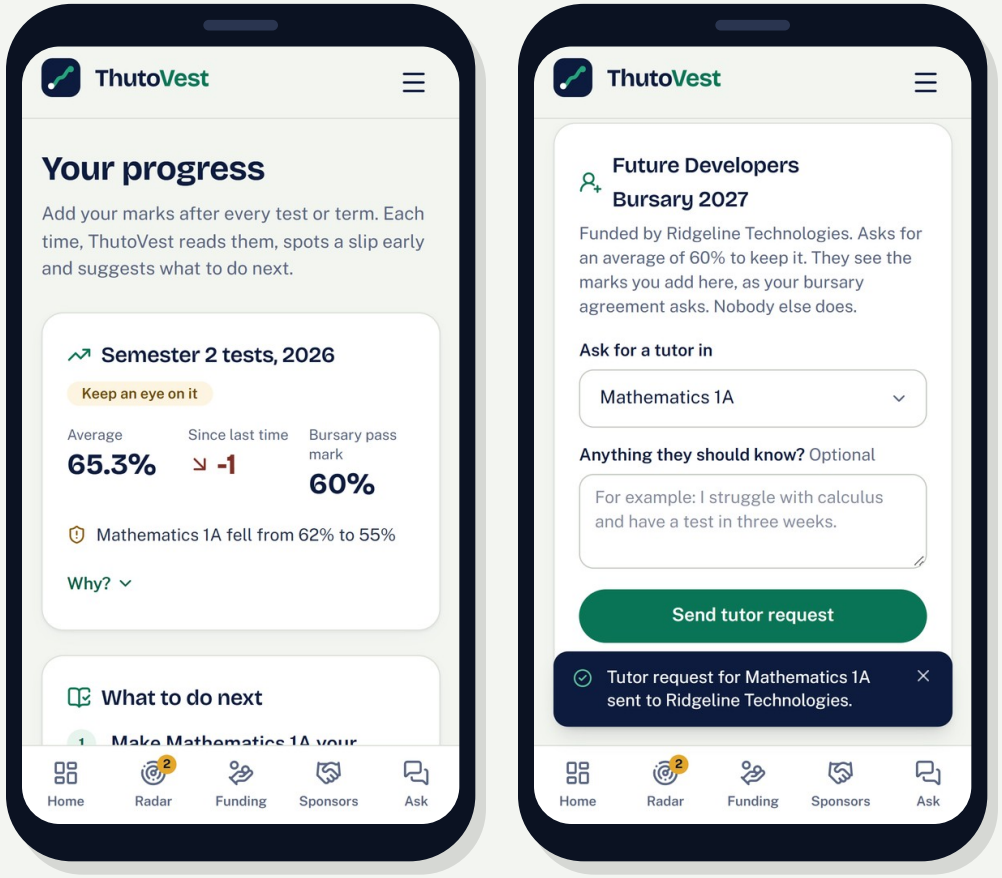
- **Under 18?** Hidden from every sponsor until a parent or guardian consents (POPIA sections 34 and 35).
- Every offer to a child needs the guardian's consent form, ID and proof of guardianship. **Nomsa co-signs.**
- **Fee requests are blocked** and the sender suspended, even in everyday words like "send me R500".
- Real sponsors never ask for money: **funding only ever goes to the institution**.



A demonstration build: the legal controls have not been reviewed by a lawyer.

STOP 4: STAY

A year later: funding is only half the job



- Every time a student adds marks, ThutoVest **reads and analyses them**: average, change, slipping modules and the bursary's pass mark.
- When Maths slips from 62% to 55%, **the assistant offers to ask the sponsor for a tutor**. One tap.
- The sponsor sees the slip early and sends a **fair warning**: the requirement, no decision yet, and real help.

"…**This is not a decision to stop your funding.** If your results don't satisfy the bursary's requirement at the next review, you could be at risk of losing this funding. Let's make a plan together."

HOW IT SUSTAINS A STUDENT

From admission to graduation, the route loops every year



Without ThutoVest

Marks slip quietly all year. A letter after final results says the bursary has stopped, when it is too late to fix.

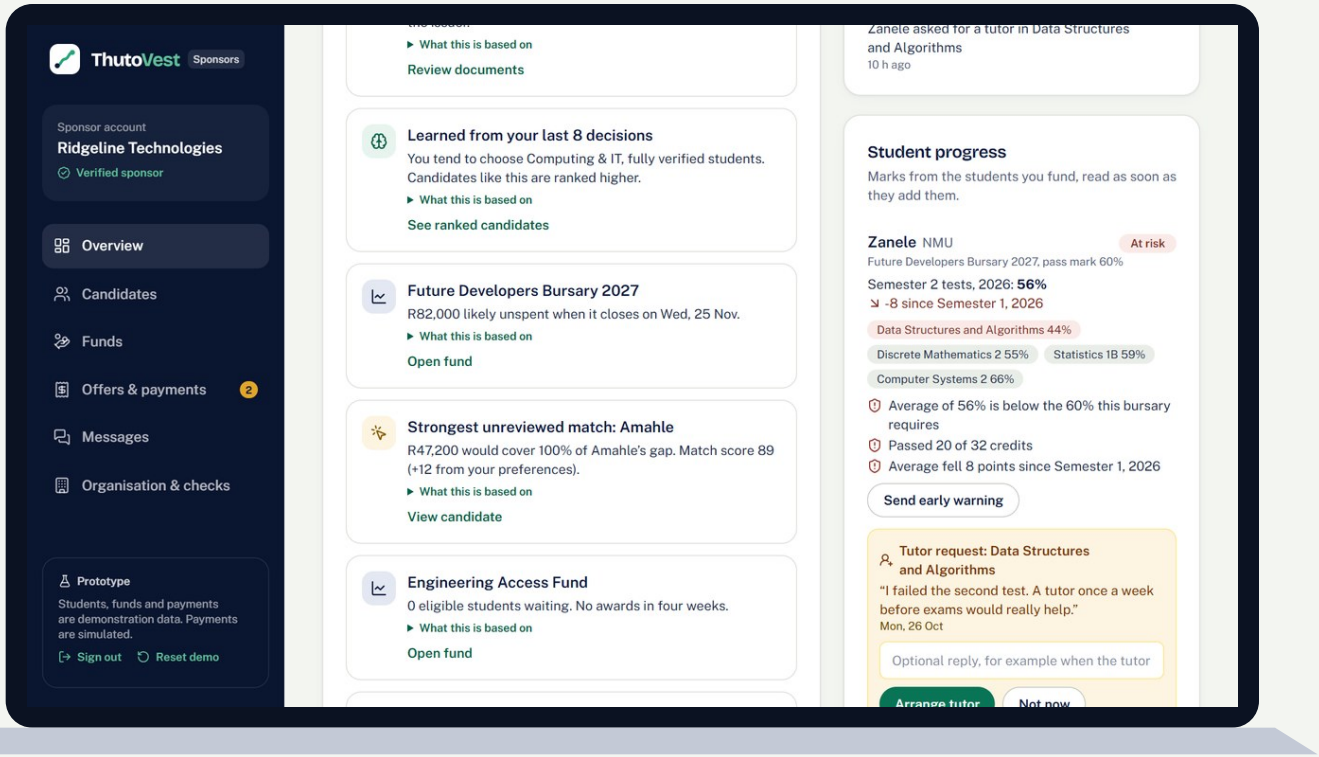
With ThutoVest

The slip shows up after the first test. The student gets a tutor, the sponsor gets a plan, and the funding is kept.

Each year: reminders to renew funding and re-register, and a second bursary if the gap grows.

FOR SPONSORS

Fund the right students, and keep them



- Candidates **verified and ranked** by fit, and by what this sponsor tends to fund.
- Funds with **their own terms**: what it pays for, a pass mark, work-back years.
- **Student progress** for everyone they fund, with tutor requests and early warnings.
- Payment locked until documents are approved, then **paid to the institution**.
- A skills-development spend estimate for **B-BBEE** reporting.

HOW IT THINKS

It notices problems before they happen

Explainable rules built on each user's own data. Every alert has a "Why?" that shows the evidence.

IT LEARNS FROM	SO IT CAN
APS, field, budget and province	Re-score every course match the moment the APS changes
Marks after every test	Spot a slipping subject, recalculate the APS, and warn before a bursary is at risk
The student's pace	Predict who will fall behind on deadlines, and nudge before they do
Costs, aid and deadlines	Show the funding gap month by month, with what-if scenarios
A sponsor's past decisions	Rank the students that sponsor is most likely to fund
Message text and documents	Block fee requests, and flag the wrong, unreadable or edited files

Honest note: explainable rules today, not machine learning yet. Machine learning comes once there is real outcome data to train on.

HOW IT IS KEPT SAFE

Security built in, and tested against the live app

99
automated checks pass on every change

38 of 38
live attack probes refused

8 of 8
database tables with row-level security

- Every action is checked on the server with the same rules code as the app, so browser and server never disagree.
- Fee and bank-detail requests are blocked and the sender suspended. Sponsors can’t verify themselves.
- Marks go only to the sponsor funding the student. Uploads are checked by their real contents.
- Signed webhooks, strict Content Security Policy, no secrets in code or history, demo logins locked.

A FEE REQUEST, STEP BY STEP

A “sponsor” writes
“Send me R500 so I can process your documents.”

›

The server reads it
The same rules as the app spot a request for money.

›

Blocked and suspended
The message never arrives. The sender can’t write again.

›

The student is told why
Real sponsors never ask for money.

HOW IT IS BUILT

One set of rules, on the phone and on the server

On the phone

- React 19, TypeScript, Vite and Tailwind
- Works on any phone or laptop
- WhatsApp bot for low data

In between

- Vercel serves the app with strict security headers
- Twilio forwards WhatsApp messages as signed webhooks
- Everything over HTTPS

On the server: Supabase

- Real logins with hashed passwords
- A platform function runs the app's own rules on every action
- Postgres with row-level security
- Live updates reach open screens in about 2 seconds

ONE TAP, END TO END

Tatenda taps Sign and accept › The platform function checks the bursary and guardian rules › Postgres saves it behind row-level security › Ridgeline's open screen updates in about 2 seconds

WHAT WE BUILT

Built during the hackathon, 23 to 26 October

The student planning core existed before the event. These were built and posted in our Sonke updates.

- **Live backend**
  Real logins, server-side rules, row-level security, live sync
- **Real institution data**
  26 universities, 50 TVET colleges, the DHET register
- **Children protected**
  Guardian consent and co-signing, enforced on the server
- **Sign-up for every stage**
  School learners and students already at university or college
- **WhatsApp bot**
  Live on real phones through Twilio
- **Fair funding rules**
  Bursary terms, two bursaries at most, no offer above the gap
- **Marks and progress**
  Analysis, advice, tutor requests and fair warnings
- **Security hardening**
  38 of 38 live attacks refused, 99 automated checks

HOW IT MAKES MONEY

Free for students, paid for by the funding it unlocks

How it earns

- Free for students, always.
- A small admin fee on awards paid through the platform, charged to the sponsor's award.
- Impact and skills-development reports for sponsors.
- Cohort insights for schools and institutions.

Where the route goes next

- Real identity and bank verification partners
- USSD menus for phones with no data
- A guardian link: copies of offers and warnings on the parent's phone
- Email, SMS and personal WhatsApp alerts
- Forecasts trained on real outcomes

HONEST ABOUT THE BUILD

What is real today, and what is simulated

Real

- Live app with real logins and live sync
- Server-side rules and row-level security
- WhatsApp bot on real phones (Twilio sandbox)
- Official university and college data, with sources
- Document checks and scam blocking
- Bursary, child-safety and progress rules on the server

Simulated, and labelled in the app

- Identity, guardian and bank verification
- Payments to institutions
- Sponsors, students and awards are fictional
- Fees and some deadlines are demo figures
- Reading marks from a report: the student checks them
- The Navigator chat is scripted

A demonstration build. The POPIA and children's controls were designed by the team from the Act and have not been reviewed by a lawyer.

QUESTIONS JUDGES ASK

What, why and how, answered

How is this different from NSFAS or a bursary site?

It joins the whole route: plan, apply, fund and stay on track, with one next step at a time.

Why would a sponsor use it?

Verified, ranked candidates, their own terms enforced, progress tracking and impact reports.

How do you stop scams?

Fee and bank requests are blocked and the sender suspended. Money only goes to institutions.

How do you protect children?

Hidden until a guardian consents; guardian documents and a co-signature on every offer.

How do students avoid losing funding?

Marks are read after every test, the assistant offers a tutor, and sponsors warn early.

How does a parent stay informed?

Today on the student's phone: consent and co-signing. Next: a guardian link.

What if a student has no data?

WhatsApp works today. USSD menus are next.

Is the intelligence real AI?

Explainable rules from each user's data, with the evidence shown. Machine learning later.

TRY IT

Ride the route yourself at thutovest.vercel.app

Use two screens: one tab as a student, one as the sponsor, side by side. Every password is Demo#2026.

Tatenda, Grade 12, 17	**tatenda@thutovest.demo**
One login per judge	**judge1 to judge5@thutovest.demo**
Nandi, funded first-year at UJ	**nandi@thutovest.demo**
Ridgeline Technologies, sponsor	**funding@ridgeline.demo**

Every account has **Reset demo** in its menu.

The three-minute route

- Tatenda: share your profile with Ridgeline
- Ridgeline: make an offer from Candidates
- Tatenda: review, sign and accept; Nomsa co-signs
- Ridgeline: approve documents, pay the institution
- Nandi: add a sample report, then ask for a tutor
- Ridgeline: Student progress, send an early warning

Every student deserves a plan, a fair chance at funding, and support until graduation.

Samalani Maluwa
Reshoketswe Makhura
Tebogo Sasa

thutovest.vercel.app | github.com/SamkiDev/thutovest

Thank you. Try it now.
thutovest.vercel.app